

# A Sample Article Using Quarto with Standard LaTeX Article Class

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**ABSTRACT:** This document demonstrates how to use Quarto with the standard LaTeX article document class. It produces PDF output suitable for preprints, working papers, and general academic writing.

**KEY WORDS:** Quarto, LaTeX, article, Pandoc, template

## 1 Introduction

中国人民 This file is a sample article demonstrating Quarto's capabilities with the standard LaTeX article document class. It supports PDF (via LaTeX) output. It is assumed that the reader has a basic working knowledge of stochastic calculus Karatzas and Shreve [1] and stochastic control theory Yong and Zhou [2].

$$\begin{aligned} \forall x \in \mathbb{R}, \quad \exists y \in \mathbb{N}: \quad \alpha + \beta = \gamma \neq \delta \leq \epsilon \approx \zeta \\ \sum_{i=1}^n x_i^2 + \prod_{j=1}^m y_j^2 = \int_a^b f(x) dx \\ \lim_{n \rightarrow \infty} \frac{1}{n} = 0, \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ S = \{x \mid x > 0\}, \quad f: X \rightarrow Y, \quad x \mapsto f(x) \\ \bigcup_{i=1}^n A_i \cap \bigcap_{j=1}^m B_j, \quad A \subseteq B, \quad A \supset B \\ \mathcal{L} = \mathcal{F}(\mathbb{R}), \quad \mathfrak{g} \cong \mathfrak{h} \\ \boxed{E = mc^2} \end{aligned}$$

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**Definition 1.1** (连续性). 设函数  $f: D \subseteq \mathbb{R} \rightarrow \mathbb{R}, x_0 \in D$ . 如果对于任意  $\epsilon > 0$ , 存在  $\delta > 0$ , 使得当  $|x - x_0| < \delta$  且  $x \in D$  时, 有  $|f(x) - f(x_0)| < \epsilon$ , 则称  $f$  在  $x_0$  处连续。

### 1.0.1 定理环境

**Theorem 1.1** (中值定理). 设函数  $f: [a, b] \rightarrow \mathbb{R}$  连续, 在  $(a, b)$  内可导. 则存在  $\xi \in (a, b)$ , 使得

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} \quad (1)$$

参考定理 Theorem 1.1 可知...

### 1.0.2 引理环境

**Lemma 1.1** (费马引理). 设函数  $f$  在  $x_0$  处可导, 由公式 (1) 知  $x_0$  是  $f$  的极值点, 则  $f'(x_0) = 0$ 。

### 1.0.3 推论环境

**Corollary 1.1** (导数零点定理). 若函数  $f$  在区间  $I$  上可导且  $f'(x) = 0$  对所有  $x \in I$  成立, 则  $f$  在  $I$  上为常数函数。

**Theorem 1.2** (Residue Theorem). *Let  $f$  be analytic in a simply connected domain  $D$ . Then for any closed contour  $\gamma \subset D$ ,*

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_k \text{Res}(f, z_k)$$

This template supports standard Quarto cross-references. You can refer to Theorem Theorem 1.2 in the text.

## 2 Markdown Basics

The template is built on Quarto and Pandoc, leveraging standard LaTeX tools for PDF output.

### 2.1 Equations

Inline math:  $E = mc^2$ . Display math:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

### 2.2 Theorems and Proofs

**Lemma 2.1** (Key Lemma). *If  $f$  is continuous on  $[a, b]$ , then  $f$  is integrable on  $[a, b]$ .*

**Corollary 2.1** (Easy Corollary). *Every continuous function on a closed interval is bounded.*

**Definition 2.1** (Model Definition). A stochastic process  $\{X_t\}_{t \geq 0}$  is a collection of random variables indexed by time.

## 2.3 Lists

1. First item
  2. Second item
  3. Third item
- Bullet one
  - Bullet two

## 2.4 Tables

Table 1: A sample table

| Variable | Description  | Unit             |
|----------|--------------|------------------|
| $x$      | Position     | m                |
| $v$      | Velocity     | m/s              |
| $a$      | Acceleration | m/s <sup>2</sup> |

## 3 Bibliography

This template uses natbib with the plainnat-doi bibliography style for author-year citations Karatzas and Shreve [1].

## Acknowledgment

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Use the `\appendix options="An Appendix"` markup if you have a single appendix. Do not use `\section{}` after `\appendix` in LaTeX.

## References

- [1] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*. Springer New York, 1991.
- [2] Jiongmin Yong and Xunyu Zhou. *Stochastic Controls: Hamiltonian Systems and HJB Equations*. Stochastic Modelling and Applied Probability. Springer New York, 1999.

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